

Proof by Mathematical Induction



The three steps of induction

- 1 State the three steps required for a valid proof by mathematical induction.
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Proving a summation formula

- 2 Prove by induction that $1 + 2 + 3 + \dots + n = n(n + 1)/2$ for all $n \geq 1$.
 - 3 Prove by induction that $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$ for all $n \geq 1$.
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Proving divisibility results

- 4 Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.
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Induction with inequalities

- 5 Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.
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Induction in calculus: derivatives of a product

- 6 Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n .
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Induction with a recursively defined sequence

- 7 A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.
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Proving a geometric-series-style summation

- 8 Prove by induction that the sum $3^0 + 3^1 + \dots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.
 - 9 Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n!$ equals $(n + 1)! - 1$ for all $n \geq 1$.
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Proving an inequality involving factorials

- 10 Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.
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Proving a divisibility result with two variables

- 11 Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.
- 12 Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.
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Induction with a recurrence relation for a sequence of sums

- 13 A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.
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Induction with a strengthened hypothesis

- 14 Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \geq 1$.
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Induction to prove a formula for the sum of cubes

- 15 Prove by induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = [n(n+1)/2]^2$ for all $n \geq 1$.
- 16 Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.
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Induction to prove a formula for a weighted geometric sum

- 17 Prove by induction that $1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n = (n-1)2^{n+1} + 2$ for all $n \geq 1$.
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A closing check: identifying an invalid induction argument

- 18 A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.
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Proving a sum-of-products formula by induction

- 19 Prove by induction that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n + 1) = n(n + 1)(n + 2)/3$ for all $n \geq 1$.
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Proving an exponential inequality by induction

- 20 Prove by induction that $2^n > n$ for all integers $n \geq 1$.
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Proving a divisibility result by induction

- 21 Prove by induction that $n^3 - n$ is divisible by 6 for all integers $n \geq 1$.
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Proving a closed form for a recursive sequence

- 22 A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for $n \geq 1$. Prove by induction that $a_n = 2^n - 1$ for all $n \geq 1$.
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A second exponential inequality by induction

- 23 Prove by induction that $3^n \geq 2n + 1$ for all integers $n \geq 1$.
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A third inequality by induction

- 24 Prove by induction that $n^2 \geq 2n - 1$ for all integers $n \geq 1$.
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Induction in combinatorics: counting subsets

- 25 Prove by induction that a set with n elements has exactly 2^n subsets, for all integers $n \geq 1$.
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Proving the sum of the first n odd numbers

26 Prove by induction that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for all $n \geq 1$.